Mathis Hardion

PhD student at the LIGM, Université Gustave Eiffel

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Mathis Hardion
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Interests

Optimal transport and its entropic regularization, gradient flows in the Wasserstein space, analysis of PDEs, geometry on the space of probability measures, applications to machine learning and sampling methods.

Education

2025 - **PhD in mathematics**
LIGM, Université Gustave Eiffel (Marne-la-Vallée, France)
Working title: *Optimization through optimal transport with entropic regularization* [1].
Supervisors: Théo Lacombe and François-Xavier Vialard.

2023 - 2024 **MSc in mathematics, vision, learning (MVA)**
École Normale Supérieure de Paris-Saclay (Gif-sur-Yvette, France)
Research-oriented degree in machine learning through a mathematical lens, wide spectrum of courses followed in the above domains of interest.
Thesis: *Gradient Flows in the Geometry of the Sinkhorn Divergence* (led to [2]).
Supervisor: Hugo Lavenant (Bocconi University).

2020 - 2024 **Engineering degree (Diplôme d'ingénieur)**
Télécom Paris (Palaiseau, France)
Specialization in Stochastic Modelling and Numerical Analysis, Signal Processing and Machine Learning.

2018 - 2020 **Classe Préparatoire au Grandes Écoles (MPSI, MP*)**
Lycée Carnot (Dijon, France)
Intensive two-year program giving rigorous training in preparation for national competitive exams allowing entry into top French graduate schools. Specialization in Mathematics, Physics and Computer Science.

Research stays

2024 (6 months) **Research Intern**
Bocconi University (Milan, Italy)
Gradient Flows in the Geometry of the Sinkhorn Divergence: derivation of the differential equation corresponding to the gradient flow of a potential energy, its main properties and long-time behavior, numerical implementation and comparison with the Wasserstein case. Entropic Optimal Transport, Gradient Flows, Functional Analysis, Riemannian Geometry, RKHS, Numerical Optimization & Visualization (Python).

Preprints

[1] M. Hardion and T. Lacombe. The Wasserstein gradient flow of the Sinkhorn divergence between Gaussian distributions, 2026. arXiv: `2602.10726 [math.AP]`. Submitted.

[2] M. Hardion and H. Lavenant. Gradient Flows of Potential Energies in the Geometry of Sinkhorn Divergences, 2025. arXiv: `2511.14278 [math.AP]`. Submitted.

Workshops

Dec. 2024 Optimal Transportation and Applications. Scuola Normale Superiore, Pisa (attendee).

Apr. 2026 Optimal transport in atmosphere and ocean problems. University of Reading, UK (remote attendee).

Talks

Feb. 2026	PhD seminar of the LIGM. Université Gustave Eiffel, Champs-sur-Marne (slides).
May. 2026	PhD seminar of the CMAP and CMLS. École Polytechnique, Palaiseau, France (slides).

Teaching

2025 - **Teaching Assistant**
Université Gustave Eiffel (Marne-la-Vallée, France)
2025-2026:
- "Labo Math-Info" (Computer science and Mathematics Lab), Year 2
- "AP2" (Algorithms and programming), Year 1

MSc research projects

Some of the reports and presentations made during my MSc can be found in the "MSc projects" section of my website, including the following:

Reports:

Neural Optimal Transport
Variational Learning of Inducing Variables in Sparse Gaussian Processes
Generalized Sliced Distances for Probability Distributions
Sparse representation of multivariate extremes with applications to anomaly detection
Mean Curvature Motion of Point Cloud Varifolds

Presentations:

Riemanniann Manifold Hamiltonian Monte Carlo
FibeRed: Fiberwise Dimensionality Reduction of Topologically Complex Data with Vector Bundles

Other work experience

2023 (2 months) **Front Office Support**
Axpo Solutions AG (Brussels, Belgium)
Constrained algorithmic financial optimization of multi-asset heat, power and CO2 production schedules for greenhouses. Applied research, Mathematical modelling, Numerical optimization (python, LP/MILP, Simulated annealing, Evolutionary algorithm), FTP communication, Predictive price curve evaluation and comparison.

2021 (2 months) **Education Intern**
Learning Robots (Gif-sur-Yvette, France)
Design and improvement of high-school and post-secondary level practical sessions and videos teaching artificial intelligence algorithms and ethics through robots. Development of new features for the AlphAI robot and software (Python).

Languages

French: Native
English: Proficient (C1)
German: Intermediate (B1)